

AMITABH CHOUDHURY

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SUMMARY

Data Engineer at Caterpillar, promoted from Systems Data Support Analyst after building domain expertise across warehouse and supply-chain data. Owns **25+ production Snowflake/dbt pipelines** serving **~75 stakeholders**; led a **2,025-line SAS-to-Snowflake migration** across 5 data domains with zero-defect cutover validated row by row. Builds rigorously evaluated data systems outside work: a **Text-to-SQL engine** with AST-based guardrails and adversarial benchmarking; a **hybrid RAG pipeline** with RAGAS evaluation and CI deployment gate; an **Iceberg/Dagster lakehouse** with 36 automated data quality checks. B.Tech AI/ML (CGPA 8.55). **LeetCode Knight** — max rating 1923, global rank 283/29,865 (top 1%).

WORK EXPERIENCE

Caterpillar

Data Engineer

Bengaluru, India

Jun 2025 – Present

- Own and operate **25+ production Snowflake pipelines** via SnapLogic and dbt serving **~75 warehouse and supply-chain stakeholders**; pipelines refresh as frequently as every **20 minutes**, processing volumes from hundreds to millions of rows daily; results surface in Power BI dashboards used for shipment-allocation decisions.
- Led technical implementation of the **MineStar MSU SAS-to-Snowflake migration**: refactored **2,025 lines of legacy SAS** into 43 Snowflake objects across 5 data domains (inventory, demand, backorders, schedules, forecasts for ~5,500 globally-sourced parts); consolidated 102 work tables to 32 via CTE-based optimization; eliminated mainframe-FTP dependencies and manual Excel reporting workflows; **validated zero-defect cutover via row-by-row parity reconciliation** across all five domains before production switch.
- Designed a Goods Receipt **dbt** pipeline with **SCD Type-2** history capture at supplier/PO grain; 13 schema and relationship tests enforce data contracts; Snowflake **Dynamic Tables** auto-refresh summary aggregates every 30 minutes.
- Identified SQL safety gaps in an internal **Snowflake Cortex** NL-to-SQL prototype — no write protection, no adversarial evaluation, no hallucination detection; rebuilt concept open-source with **AST-based guardrails** and a 50-question adversarial benchmark — see *Text-to-SQL* project.
- Built a **Streamlit** operations app for the Tidal monitoring team supporting **24/7 shift handovers**; categorises critical jobs by shift, surfaces runbook-style remediation steps, persists incident metadata in a Snowflake audit table, and auto-generates shift-transition emails.

Caterpillar

Systems Data Support Analyst

Bengaluru, India

Jun 2024 – Jun 2025

- Delivered ad hoc analytical SQL across Snowflake for warehouse, procurement, and finance stakeholders; translated operational and supply-chain data into insights that fed procurement-planning discussions.
- Initiated a multi-year warehouse parts usage analysis using **Python and pandas** to surface seasonal demand patterns; built a **Random Forest** demand-forecasting prototype with time-based train/test split — findings carried forward into the Data Engineer role.
- Supported and extended production **Snowflake/SnapLogic/Tidal** pipelines alongside senior engineers; handled data-quality triage and validation to ensure reliable refreshes for downstream analytics consumers.

Caterpillar

Software Engineer Intern

Bengaluru, India

Jun 2023 – Sep 2023

- Implemented **KS-test + PSI** drift detection on 3 production Tidal data streams; tuned alert thresholds against 6 months of historical data, reducing false-positive alerts by ~30% and cutting unnecessary pipeline re-runs escalated to the data quality team.

SKILLS

Languages: Python, SQL, C++

Data Engineering: Snowflake (Dynamic Tables, Tasks, Streams, Cortex), dbt (SCD Type-2, snapshots, tests), Apache Iceberg, DuckDB, Dagster, Soda Core, SnapLogic, Tidal

Evaluation & GenAI: RAGAS, Langfuse, sentence-transformers (BGE-large, MiniLM-L6), Qdrant, BM25/RRF, FastAPI, instructor, Pydantic v2, sqlparse, SQLAlchemy

Analytics & BI: Pandas, Power BI, scikit-learn

Infra & Dev: Docker, GitHub Actions, Git, PostgreSQL 15

PROJECTS

GitHub Archive Lakehouse | [GitHub](#)

Apache Iceberg · DuckDB · Dagster · Soda Core · MinIO · Streamlit · Docker

- Built a Bronze/Silver/Gold **Iceberg lakehouse** ingesting **7.4M+ GitHub Archive events**; manifest-tracked, **crash-resumable pipeline** handles partial MinIO writes and mid-run failures; Silver layer deduplicates by event ID, parses 15 event-type payloads into typed columns, and routes malformed rows to a quarantine table with error reason.
- **36 Soda Core** data quality checks across all three layers gate each Dagster run; Software-Defined Assets with DailyPartitionsDefinition provide full asset lineage; **Iceberg time-travel and live schema evolution** demonstrated on a 3.8M-row table; 5 documented failure scenarios with exact recovery commands.

Text-to-SQL with Guardrails & Hallucination Detection | [GitHub](#)

FastAPI · PostgreSQL 15 · Claude Sonnet · instructor · Pydantic v2 · sqlparse · sentence-transformers · Docker

- NL-to-SQL system with a multi-layer safety stack: **sqlparse AST guardrails** (token-type checking, not regex) hard-block DDL/DML, multi-statement injection, and catalog access; all execution runs inside **PostgreSQL SET TRANSACTION READ ONLY** as a database-enforced fallback independent of application logic.
- Evaluated on a **50-question hand-curated adversarial benchmark** (JOINS, CTEs, date arithmetic, ranking); passed all executable queries; blocked all 10 seeded SQL injection/unauthorized write tests; documented edge-case failures for unanswerable queries; **back-translation hallucination detection** via MiniLM-L6 cosine similarity; 6 Architecture Decision Records capture design trade-offs.

Evaluated RAG for Snowflake & dbt Docs | [GitHub](#) | [Live Demo](#)

BGE-large-en-v1.5 · BM25/RRF · Qdrant · RAGAS · Claude Sonnet · Langfuse · GitHub Actions · Docker · Render

- Hybrid retrieval pipeline (**dense BGE-large + BM25 via RRF k=60**) with cross-encoder reranking over crawled Snowflake docs, dbt docs, and 200 dbt Discourse threads; Claude Haiku query rewriter expands terse questions before retrieval; answers grounded with cited source chunks; live on Render.
- 50-question adversarial benchmark across 4 tiers with precision traps targeting Snowflake/dbt misconceptions; **CI eval gate in GitHub Actions fails the build on RAGAS faithfulness regression**; 3-config ablation study quantifies contribution of reranking vs. hybrid retrieval; **Langfuse** traces all retrievals.

ACHIEVEMENTS

- **LeetCode Knight** — max rating **1923**; global rank **283 / 29,865** in Weekly Contest 366 (top 1%) | [amitabhchoudhury19999](#)
- **Codeforces** — max rating **1800** (Expert) | [amitabh1208](#)
- **CodeChef** — max rating **1711** (4*) | [amitabh1208](#)
- **Centurion Hackathon (National Level)** — Team Lead; placed **13th** out of 500+ participants.

EDUCATION

M.S Ramaiah University of Applied Sciences, Bengaluru — B.Tech in Artificial Intelligence and Machine Learning, CGPA 8.55/10 *Apr 2024*

Coursework: Machine Learning I & II, Deep Learning, NLP, Pattern Recognition, DBMS, Data Structures, OS, Networks.